"Sınav esnasında çözümlerini başkalarıyla paylaşmanın kopya vermek başkalarından veya dijital ortamdan yardım almanın kopya çekmek olarak yorumlanacağı konusunda sınav gözetmeni tarafından bilgilendirildim. Teslim edeceğim bu çalışmanın benim kendi çalışmam olacağını ve hiçbir şekilde kopya çekmeyeceğini kabul ve taahhüt ederim"

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29.01.2021

# Soru 1

Take the two matrices with the nomber assigned to you and aswer the question. Your matrix M and its reduced row eshelon form R are given below

a) Find basis vectors for the column space, the row space, and the null space.

b) Solve $Mx=0$ in terms of free veriables where $X=[x_1, x_2, x_3, x_4, x_5, x_6]^T$

$$M_{158} = \begin{bmatrix} -2 & -1 & -1 & -2 & 2 & 4 \\ 2 & -1 & -1 & 6 & -4 & -2 \\ -2 & 3 & 1 & -8 & 4 & -4 \\ 2 & 5 & 3 & -4 & 1 & -9 \end{bmatrix} \qquad R_{158} = \begin{bmatrix} 1 & 0 & 0 & 2 & 0 & 3 \\ 0 & 1 & 0 & -1 & 0 & -3 \\ 0 & 0 & 1 & -1 & 0 & -1 \\ 0 & 0 & 0 & 0 & 1 & 3 \end{bmatrix}$$

$$\text{Column Space} = \alpha\begin{bmatrix} -2 \\ 2 \\ -2 \\ 2 \end{bmatrix} + \beta\begin{bmatrix} -1 \\ -1 \\ 3 \\ 5 \end{bmatrix} + \theta\begin{bmatrix} -1 \\ -1 \\ 1 \\ 3 \end{bmatrix} + \gamma\begin{bmatrix} 2 \\ -4 \\ 4 \\ 1 \end{bmatrix}$$

$\dim(\text{col}) = 4$

$$\text{row space} = \alpha\begin{bmatrix} 1 \\ 0 \\ 0 \\ 2 \\ 0 \\ 3 \end{bmatrix} + \beta\begin{bmatrix} 0 \\ 1 \\ 0 \\ -1 \\ 0 \\ -3 \end{bmatrix} + \theta\begin{bmatrix} 0 \\ 0 \\ 1 \\ -1 \\ 0 \\ -1 \end{bmatrix} + \gamma\begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 3 \end{bmatrix}$$

$\dim(\text{row}) = 4$

$$\text{null space} = x_4\begin{bmatrix} -2 \\ 1 \\ 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_6\begin{bmatrix} -3 \\ 3 \\ 1 \\ 0 \\ -3 \\ 1 \end{bmatrix}$$

$\dim(\text{null}) = 2$

$$\begin{matrix} x_1 & x_2 & x_3 & x_4 & x_5 & x_6 \end{matrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 2 & 0 & 3 \\ 0 & 1 & 0 & -1 & 0 & -3 \\ 0 & 0 & 1 & -1 & 0 & -1 \\ 0 & 0 & 0 & 0 & 1 & 3 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{bmatrix} = 0$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{bmatrix} = \begin{bmatrix} -2x_4 - 3x_6 \\ x_4 + 3x_6 \\ x_4 + x_6 \\ x_4 \\ -3x_6 \\ x_6 \end{bmatrix}$$

$x_4$ and $x_6$ free veriables.

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# Soru2

Take the vector v with the number assing to you and answer the question.

a) Show that the vectors $u_1, u_2, u_3, u_4$ are orthogonal.

b) Express your vector v in terms of the orthogonal vectors $u_1, u_2, u_3, u_4$.

Hint: Express $V = a_1 u_1 + a_2 \cdot u_2 + a_3 \cdot u_3 + a_4 \cdot u_4$ $\quad a_1 = \frac{v \cdot u_1}{u_1 \cdot u_1} = \frac{(v, u_1)}{(u_1, u_1)}$

$$u_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}, \quad u_2 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ -1 \end{bmatrix}, \quad u_3 = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix}, \quad u_4 = \begin{bmatrix} 0 \\ 1 \\ -1 \\ 0 \end{bmatrix}$$

$$V158 = \begin{bmatrix} 7 \\ 8 \\ 2 \\ 7 \end{bmatrix}$$

a) $u_1 \cdot u_2 \overset{?}{=} 0$

$\begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \\ 0 \\ -1 \end{bmatrix} = 0$ ortogonal

$u_1 \cdot u_3 \overset{?}{=} 0$

$\begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix} = 0$ ortogonal

$u_1 \cdot u_4 \overset{?}{=} 0$

$\begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1 \\ -1 \\ 0 \end{bmatrix} = 0$ ortogonal

$u_2 \cdot u_3 \overset{?}{=} 0$

$\begin{bmatrix} 1 \\ 0 \\ 0 \\ -1 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix} = 0$ ortogonal

$u_2 \cdot u_4 \overset{?}{=} 0$

$\begin{bmatrix} 1 \\ 0 \\ 0 \\ -1 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1 \\ -1 \\ 0 \end{bmatrix} = 0$ ortogonal

$u_3 \cdot u_4 \overset{?}{=} 0$

$\begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1 \\ -1 \\ 0 \end{bmatrix} = 0$ ortogonal

b) $a_1 = \frac{V158 \cdot u_1}{u_1 \cdot u_1} = \frac{[7827] \cdot \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}}{[1001] \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}} = \frac{14}{2} = 7$

$a_2 = \frac{V158 \cdot u_2}{u_2 \cdot u_2} = \frac{[7827] \cdot \begin{bmatrix} 1 \\ 0 \\ 0 \\ -1 \end{bmatrix}}{[100-1] \begin{bmatrix} 1 \\ 0 \\ 0 \\ -1 \end{bmatrix}} = \frac{0}{2} = 0$ olduğu için yoktur.

$a_3 = \frac{V158 \cdot u_3}{u_3 \cdot u_3} = \frac{[7827] \cdot \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix}}{[0110] \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix}} = \frac{10}{2} = 5$

$a_4 = \frac{V158 \cdot u_4}{u_4 \cdot u_4} = \frac{[7827] \begin{bmatrix} 0 \\ 1 \\ -1 \\ 0 \end{bmatrix}}{[01-10] \begin{bmatrix} 0 \\ 1 \\ -1 \\ 0 \end{bmatrix}} = \frac{6}{2} = 3$

$V = 7u_1 + 5u_3 + 3u_4$

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# Soru3

Find the eigenvalues $\lambda$ and eigenvectors $v$ of your matrix.

$$M158 = \begin{bmatrix} 4 & 4 & 6 \\ 0 & 6 & 3 \\ 0 & 0 & 5 \end{bmatrix}$$

Find the eigenvalues;

$$\begin{bmatrix} 4-\lambda & 4 & 6 \\ 0 & 6-\lambda & 3 \\ 0 & 0 & 5-\lambda \end{bmatrix} = (4-\lambda)\cdot(6-\lambda)\cdot(5-\lambda) = 0$$

$\downarrow$ 0

$\lambda_1 = 4 \qquad \lambda_2 = 6 \qquad \lambda_3 = 5$

Find the eigenvectors;

$A - \lambda_i \cdot I$

## $\lambda_1 = 4$

$$M158 = \begin{bmatrix} 4 & 4 & 6 \\ 0 & 6 & 3 \\ 0 & 0 & 5 \end{bmatrix}$$

$$A - \lambda_1 I = \begin{matrix} x_1 & x_2 & x_3 \end{matrix} \begin{bmatrix} 0 & 4 & 6 \\ 0 & 2 & 3 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$2x_2 + 3x_3 = 0$$

$$x_3 = 0$$
$$x_2 = 0$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

## $\lambda_2 = 6$

$$A - \lambda_2 I = \begin{matrix} x_1 & x_2 & x_3 \end{matrix} \begin{bmatrix} -2 & 4 & 6 \\ 0 & 0 & 3 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$-2x_1 + 4x_2 + 6x_3 = 0$$

$\downarrow$

$$x_3 = 0$$

$$-2x_1 = -4x_2$$
$$x_1 = 2x_2$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_2 \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$$

## $\lambda_3 = 5$

$$A - \lambda_3 I = \begin{matrix} x_1 & x_2 & x_3 \end{matrix} \begin{bmatrix} -1 & 4 & 6 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$-x_1 + 4x_2 + 6x_3 = 0$$

$$x_2 + 3x_3 = 0$$

$\downarrow$

$$x_2 = -3x_3$$

$$-x_1 - 12x_3 + 6x_3 = 0$$

$$x_1 = -6x_3$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} -6 \\ -3 \\ 1 \end{bmatrix}$$

$$V = \left\{ x_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, x_2 \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}, x_3 \begin{bmatrix} -6 \\ -3 \\ 1 \end{bmatrix} \right\} \quad ?$$

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# Soru 4

Apply Gram Schmidt ortogonalization to your three vectors.

Vectors 158

$$\begin{bmatrix}0\\2\\4\\2\end{bmatrix}\begin{bmatrix}-1\\0\\-4\\-4\end{bmatrix}\begin{bmatrix}0\\-4\\4\\-4\end{bmatrix}$$

$x_1$ $x_2$ $x_3$

$x_1 = v_1$

$$v_1 = \begin{bmatrix}0\\2\\4\\2\end{bmatrix}$$

$$v_2 = x_2 - \frac{x_2 \cdot v_1}{v_1 \cdot v_1} \cdot v_1$$

$$= \begin{bmatrix}-1\\0\\-4\\-4\end{bmatrix} - \frac{[-1\ 0\ -4\ -4]\begin{bmatrix}0\\2\\4\\2\end{bmatrix}}{[0\ 2\ 4\ 2]\begin{bmatrix}0\\2\\4\\2\end{bmatrix}} \cdot \begin{bmatrix}0\\2\\4\\2\end{bmatrix}$$

$$\begin{bmatrix}-1\\0\\-4\\-4\end{bmatrix} - \frac{(-24)}{24} \cdot \begin{bmatrix}0\\2\\4\\2\end{bmatrix}$$

$$= \begin{bmatrix}-1\\0\\-4\\-4\end{bmatrix} + \begin{bmatrix}0\\2\\4\\2\end{bmatrix}$$

$$v_2 = \begin{bmatrix}-1\\2\\0\\-2\end{bmatrix}$$

$$v_3 = x_3 - \frac{x_3 \cdot v_1}{v_1 \cdot v_1} \cdot v_1 - \frac{x_3 \cdot v_2}{v_2 \cdot v_2} \cdot v_2$$

$$= \begin{bmatrix}0\\-4\\4\\-4\end{bmatrix} - \frac{[0\ -4\ 4\ -4]\begin{bmatrix}0\\2\\4\\2\end{bmatrix}}{[0\ 2\ 4\ 2]\begin{bmatrix}0\\2\\4\\2\end{bmatrix}} \cdot \begin{bmatrix}0\\2\\4\\2\end{bmatrix} - \frac{[0\ -4\ 4\ -4]\begin{bmatrix}-1\\2\\0\\-2\end{bmatrix}}{[-1\ 2\ 0\ -2]\begin{bmatrix}-1\\2\\0\\-2\end{bmatrix}} \cdot \begin{bmatrix}-1\\2\\0\\-2\end{bmatrix}$$

$$= \begin{bmatrix}0\\-4\\4\\-4\end{bmatrix} - \frac{0}{24} \cdot \begin{bmatrix}0\\2\\4\\2\end{bmatrix} - \frac{0}{1} \begin{bmatrix}-1\\2\\0\\-2\end{bmatrix}$$

$$v_3 = \begin{bmatrix}0\\-4\\4\\-4\end{bmatrix}$$

ortogonal baz $= \left\{ \begin{bmatrix}0\\2\\4\\2\end{bmatrix}, \begin{bmatrix}-1\\2\\0\\-2\end{bmatrix}, \begin{bmatrix}0\\-4\\4\\-4\end{bmatrix} \right\}$